Below the Noise Floor: A Deterministic GLEE Agent and the Measurement Problem It Exposed

Yosef Zoabi
Faculty of Data and Decision Sciences
Technion – Israel Institute of Technology, Haifa, Israel
yosef.zoabi@campus.technion.ac.il
GLEE Agent ID edb1db91303e (7aidara_Gamma)
6th agent track, 2nd human track. github.com/YosefZoabi/glee-agent

Abstract

I study the difficulty of evaluating small strategy changes in a live economic-game competition through a deterministic, LLM-free agent. The agent placed sixth in the GLEE agent track; its development fleet logged 264,432 games across bargaining, negotiation and persuasion. In a simultaneous 8.6-hour comparison, five slots deployed from the same reference build each played about 3,000 games. They began 85 overall rating points apart because their rating histories differed, so the 88-point maximum spread largely reflects that inherited gap. They ended 46 points apart, and the slot that started last finished first. Under a fitted rating model and assumed game counts, only 6–21 points of the starting gap would persist, which suggests that most of the terminal spread accrued during the window; matchmaking, sampling and changes in the reference pool remain unresolved. Decision-level replay and terminal acceptance counterfactuals help investigate the behavior behind these scores: some interventions changed no actions, while transcript inspection exposed message-parser and termination-limit failures. An observational comparison with my second-place human-track play shows different participation behavior, but does not isolate message-understanding ability. The results motivate simultaneous controls, explicit rating-model assumptions and behavioral audits when evaluating small interventions during short competition windows. Code and analysis scripts are public; full empirical reproduction depends on the underlying game records.

1 Introduction

GLEE [10] evaluates strategic interaction in bargaining, negotiation and persuasion. Its competition provides a setting in which to ask a practical question: *how can an entrant distinguish a useful policy change from a noisy score change?* I address this through 7AIDARA_GAMMA, a deterministic agent that finished sixth with an overall rating of 2360.8, and through my own human-track play, which finished second.

The contribution is an empirical case study of evaluation and behavior rather than a new equilibrium solution. Automated-negotiation research already studies agent evaluation, modular strategies and acceptance conditions [1, 3, 2]; uncertainty in benchmark comparisons is also an established concern [11]. Here, five simultaneous identical-policy controls reveal rating dispersion in a live competition, and offline audits distinguish interventions that changed logged decisions from those that left every decision unchanged. Failed experiments and implementation errors are part of the evidence, not merely development history.

40th Conference on Neural Information Processing Systems (NeurIPS 2026). Workshop: Interpreting Agent Behavior (IAB) – GLEE Competition Track.

2 The agent and its operating assumptions

A dispatcher routes games to three Python policies (858 / 660 / 781 lines), configured by 114 tunables; its development fleet logged 264,432 games (Appendix Table A2). Parameters are fixed during a run; within-game state and belief updates are deterministic given the observed history. Seller randomization is seeded from the game ID and round. A safety layer repairs invalid actions and supplies a legal fallback after an exception. The reference build is covered by 472 behavioral tests.[1]

Bargaining. The policy uses the infinite-horizon proposer-share anchor $(1 - \delta_{\text{them}})/(1 - \delta_{\text{us}}\delta_{\text{them}})$ [9], with empirical floors, concession rules and explicit endgame handling. This is a heuristic benchmark, not a best-response guarantee against arbitrary opponents. Finite-horizon equilibrium predictions depend on the remaining horizon and seat; the agent instead exploits the empirically favorable final-proposer position (91.9% agreement in a 1,415-game cohort), while accounting for termination risk.

Negotiation. Observed values came from $(80, 100, 120, 150)$ scaled by $\{1, 10^2, 10^4\}$, with both players on the same scale. The policy therefore prices at possible tradable rungs. Independent uniform draws on that ladder predict positive surplus in $6/16 = 37.5\%$ of value pairs; independence of roles alone would not imply that rate. For a seller holding the top rung, or a buyer holding the bottom one, no counterparty value permits a strictly profitable trade. The agent can see this from its own value alone, so in such an *untradable seat* it names an out-of-pool price on a proposal turn and walks away at its next decision turn.

Persuasion. In each cell a buyer values a high-quality unit at m times its price and a low-quality unit at nothing, and p is the prior probability of high quality, so buying blind has positive expected value exactly when $m \cdot p > 1$. Consistent with GLEE's documented payoff structure, the competition seller received the full price on a sale regardless of quality (checked over 1,829 sold rounds). For a buyer who trusts a known recommendation policy and purchases at posterior threshold τ, recommending every high unit and a fraction q of low ones gives $\Pr(H \mid \text{recommend}) = p/[p + (1-p)q]$ and $q_{\max} = \min\{1, p(1-\tau)/[\tau(1-p)]\}$. Because GLEE lacks commitment, this Bayesian-persuasion threshold [6] is a heuristic starting point, not a guarantee of optimality. The seller ramps its budget with the horizon. The buyer combines message parsing with within-game purchase outcomes and recommendation frequency, and vetoes message templates that have only delivered low quality (Appendix Table A1); quality is hidden after a decline.

3 Identical policies, different leaderboard histories

Because only an account's best slot is ranked, I spent five slots as simultaneous controls on the final full day of the competition. All five were deployed from the same reference build (Appendix B describes how far that identity was verified) and were polled together 45 times over 8.6 hours while each played 2,945–3,195 games. Their rating states were not reset: they started between 2177 and 2263, an 85.3-point overall spread, and ended between 2299 and 2344, a 45.8-point spread. The overall spread averaged 48.1 points, and its 88.0-point maximum came ten minutes into the window, so the maximum reflects the inherited gap rather than divergence. Measured from each slot's own start (Figure 1b), the five gained between $+38.8$ and $+166.9$ points, so their gains ended 128.1 points apart after peaking near 135; the slot that started lowest finished highest, every slot led at least one snapshot, and the 45 snapshots showed 31 distinct orderings of the five. Within single families the spread reached 292.9 points. The polls recorded each slot's family ratings and total game counter; I did not retain per-family or per-role game counts.

Competition scoring compares payoffs within a configuration and role and adjusts for opponent strength, and the overall rating is the unweighted mean of the three family ratings, which I verified against the leaderboard API. To interpret updates I use a simplified per-family EMA, $R_{t+1} = (1-\eta)R_t + \eta g_t$, with $\eta \approx 0.002$ inferred from rating deltas and g_t an adjusted game target on the rating scale. This is an approximation I fitted, not the documented server rule, and every conclusion that relies on it is model-dependent. At constant η an offset decays by $(1-\eta)^n$ after n family games, a half-life of about 346 games. Under independent, stationary targets the effective sample size is

[1] **LLM usage.** The competition agent calls no language model. I used LLM coding assistance for analysis scripts and for drafting and revising the paper.

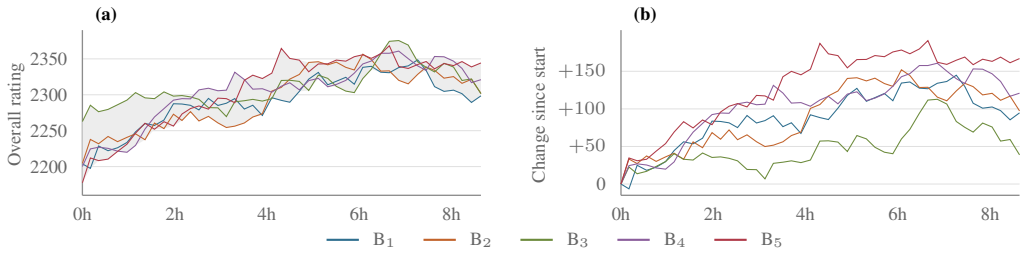


Figure 1: Five slots deployed from the same reference build, polled together 45 times over 8.6 hours. (a) Overall rating; shading is the observed min–max envelope, not an uncertainty interval. Rating states were not reset: B_1–B_5 started at 2203.9, 2204.5, 2262.7, 2200.1 and 2177.4 (spread 85.3) and ended at 2298.5, 2301.8, 2301.5, 2321.0 and 2344.3 (spread 45.8). (b) Change since each slot's first snapshot, $R_i(t) - R_i(0)$.

$(2-\eta)/\eta = 999$; lifetime game volume therefore need not describe the information in the current rating.

Applied to Figure 1, the model separates inherited from accrued dispersion. With $\bar{R}(0)$ the five slots' mean starting rating and $\lambda = (1-\eta)^n$ the share of an offset that survives n family games, I compute each slot's adjusted final rating $R_i(T) - \lambda\,[R_i(0) - \bar{R}(0)]$. Taking about 1,000 games per family as an illustrative equal split and varying it from 700 to 1,300, λ falls from 25% to 7%: 6–21 points of the 85.3-point starting gap would persist, and the adjusted final ratings span 49–64 points, so most terminal dispersion accrued during the window. These are scenarios rather than observed exposures, and the adjusted and observed spreads are not additive components. Varying η from 0.0015 to 0.0025, half-lives of 462 to 277 games, widens these ranges to 3–30 and 47–73 points without changing that conclusion. This decomposition depends on the rating model, the assumed split and rounded plotted values, and within the accrued part matchmaking, sampling and pool changes remain jointly confounded. All of these spreads describe dependent snapshots; they are neither confidence intervals nor an identified noise floor.

The reference pool changes as games enter it, making unmatched cross-time comparisons confounded. Over this window the five slots also rose by 104 points on average without any policy change, so an unmatched before-and-after comparison would have credited about 100 points to whatever changed at its start. Selecting or freezing a high-scoring slot also makes stopping time part of the evaluation protocol. A further distinction concerns the outcome being tested: for consecutive family updates, $\overline{\Delta R} = (R_{\text{end}} - R_{\text{start}})/n$. Mean rating change therefore tends toward zero as a rating settles, even for a strong policy. An IID calculation based on such deltas cannot by itself establish a sample requirement for detecting a difference in policy strength (Appendix D).

4 Behavioral measurements beyond the displayed rating

Terminal acceptance counterfactuals. An available offer can be evaluated exactly at its decision point: accepting a share s in round r banks $s\,\delta^{r-1}$. Comparing that value with the realized continuation made 23,617 refusals informative about acceptance rules, including continuations ending without agreement. The comparison is local and retrospective: it does not value every alternative policy or prove ex-ante optimality, and refusals within one game share an outcome.

Decision-level replay. The reference policy reproduced 4,272 of 4,272 bargaining and 13,056 of 13,056 negotiation decisions in the audited cohorts. Replay then counted the actions a candidate flag would change, rather than how often its condition held. A known-zone floor fired on 43% of eligible negotiation games yet changed 0 of 13,056 decisions, and hard-regime rationing changed 0 of 13,479 seller decisions. Both had already consumed a live window. Counting changed actions takes minutes and should precede any live arm. Replay validates the implementation on logged states; it does not estimate a policy's value, because a changed action alters the states that follow.

5 Interventions and implementation failures

Across 30+ interventions and 49 runs, I obtained the clearest diagnostic evidence for implementation repairs and structural policy changes (Appendix Table A4); Table A5 records these alongside the null, abandoned and retracted interventions. Their evidence types differ; none of the empirical comparisons alone establishes game-theoretic dominance. From 21 August my shipping rule required a standardized effect of 3 or two agreeing rounds against simultaneous controls, a rule the final-morning change below broke; hypotheses and cell inclusion rules were not fixed before outcomes were inspected, so all reported statistics are exploratory. A standardized effect here is an estimated difference divided by its estimated standard error on the unit listed for it, without adjustment for dependence or multiple comparisons. Seven consecutive tuning arms yielded no reliable improvement in the comparisons used. A seller arm that raised the rate of recommending low-quality units to 100% sold 11.13 units per game against 10.64 for controls at about 40%. Rating did not follow: regressing per-game rating change on push rate across 29 agent-runs, one observation per slot per run with both centred within each run, gave a slope of -0.048 rating points per game per 100 percentage points of push (OLS s.e. 0.162, which ignores dependence among slots in a run; $t = -0.29$; 95% interval -0.37 to $+0.27$). The interval admits a small gain or loss, so this does not establish equivalence.

Transcript inspection found purchases following messages such as "Not this one… pass this round." The parser recognized "not this round" but not "not this one" and interpreted unmatched text as a recommendation. Re-scoring 123,700 buyer rounds reclassified 270 purchases, 258 of them low quality; this reclassifies recorded purchases and does not estimate what deploying the corrected parser earned. Another transcript exposed an acceptance gate opening at round 97 in open-horizon games that the server ended at round 96. A separate message intervention changed take-up from 0.689 to 0.693; that observation does not establish that language generally has no effect.

On the final morning, a fleet-wide concession change was followed by drops of 100–385 negotiation rating points over five hours. With no concurrent control I cannot attribute the full decline to the parameter; the operational lesson is to keep one. Abandonment was also clustered: 80 of 109,881 rated games, with 78 in seven clock hours. Their recorded deltas averaged -6.76 and summed to about -541; the sum overstates their persistent effect on an EMA rating (Appendix D).

6 Agent Behavior Analysis

Observed behavior. Bargaining settled at median round 2, with 1.5% no-deals and mean banked share 0.429. In one slot's logs after the walk-away rule shipped (26–30 August), negotiation walked away in all 1,380 untradable games that reached its decision turn. Another 324 ended earlier, giving an overall rate of 81.0% with no observed departure from the rule. Negotiation agreed in 20–24% of hidden-value games versus 85–89% with complete information. The persuasion buyer bought 17.4 units per game where buying blind had positive expected value ($m \cdot p > 1$), versus 2.2 where it did not. These are sample descriptions, not controlled treatment effects.

Design alignment and failures. The 472 tests encode intended actions in named regimes. The reference snapshot passed all 472; a different software state, the final-night working tree carrying experimental flags, failed 27. Together with replay, these tests check implementation consistency rather than utility optimality. Four recurring deviations were unreachable rules, a gate keyed to the wrong termination limit, a permissive parser default and excessive last-round concession. In one bounded negotiation cohort, 403 of 469 seller games reached a round-10 agreement, banking a mean 0.0433 of buyer value (Appendix Table A6). Agreement rate alone concealed that behavior because capitulation still counted as a deal.

7 An observational same-author human baseline

I played 2,690 rated human-track games and finished second. Sharing the game families and opponent pool makes a useful case study, but does not match exact opponents, messages or times. Table 1 compares shared bargaining cells and pooled persuasion regimes. Bargaining equally weights retained cells matched on horizon type, seat and discount factors; persuasion pools by $m \cdot p$, so within-regime configuration composition may still differ.

Table 1: Observational human–agent comparison. Bargaining: mean banked share. Persuasion: payoff per game in product-price units; t has Welch's standard-error form, signed human minus agent, without dependence adjustment. Persuasion rows contain 48–92 human and 4,729–9,467 agent games.

Bargaining (known horizon)				**Persuasion (buyer)**			
seat	cells	human	agent	regime	human	agent	t
player 1	14	0.444	0.447	$m \cdot p > 1$	$+24.22$	$+22.10$	$+1.31$
player 2	13	0.512	0.720	$m \cdot p = 1$	$+2.99$	$+2.56$	$+0.56$
				$m \cdot p < 1$	-0.61	$+0.18$	-2.44

Among purchased items, the human's high-quality rate was 0.738 over 2,745 purchases and the agent's was 0.725 over 59,603. These rates concern selected purchases, not message-reading accuracy. The agent's higher bargaining share in the known-horizon second seat, and my negative mean persuasion payoff where $m \cdot p < 1$, are consistent with different participation and acceptance behavior. They do not identify discipline rather than perception as a causal mechanism, nor support a general claim about humans versus machines.

8 Limitations and conclusion

This is one account, one human participant and a changing competition pool. The identical-policy experiment did not reset ratings or isolate matchmaking, and its split into inherited and accrued dispersion rests on a fitted rating model. Pooled tests do not remove all dependence, and repeated tuning with flexible cell filters raises the risk of selective findings. The repository provides policies, tests and analysis scripts, but full empirical reproduction needs private game records and deployment metadata that organizer data alone would not supply.

Lessons for entrants. Five practices would have saved most of my wasted windows. (1) Replay a candidate change over logged states and count the actions it changes before giving it a live window. (2) Run at least three simultaneous controls: with two, one arm read as a standardized effect of $+2.1$ until a third control erased it. (3) Judge a change by outcomes in the cells it can affect, not by rating movement. (4) Before trusting a rate, ask what a game must do to enter its denominator; three of my measurements excluded by construction the cases that would have contradicted them. (5) Never pool games across a change; each version needs its own window.

Suggestions for live competitions. Four changes would make a live leaderboard more informative: publishing the exact rating update; reporting each rating with an uncertainty or an effective game count; letting an entrant start a control slot from a common rating; and exposing per-family game counts and scores against a fixed reference pool.

The finding is not that leaderboards fail. It is that slots deployed from one reference build ended tens of rating points apart, with most of that gap opening within a single window under the fitted model, while behavioral audits exposed failures that score summaries hid. When experimental windows are short, simultaneous controls, explicit estimands, replay and transcript inspection supply evidence that a rating cannot.

References

[1] T. Baarslag, K. Fujita, E. H. Gerding, K. Hindriks, T. Ito, N. R. Jennings, C. Jonker, S. Kraus, R. Lin, V. Robu, and C. R. Williams. Evaluating practical negotiating agents: results and analysis of the 2011 international competition. *Artificial Intelligence*, 198:73–103, 2013.

[2] T. Baarslag, K. Hindriks, and C. Jonker. Acceptance conditions in automated negotiation. In *Complex Automated Negotiations: Theories, Models, and Software Competitions*, volume 435 of *Studies in Computational Intelligence*, pages 95–111. Springer, 2013.

[3] T. Baarslag, K. Hindriks, M. Hendrikx, A. Dirkzwager, and C. Jonker. Decoupling negotiating agents to explore the space of negotiation strategies. In *Novel Insights in Agent-based Complex Automated Negotiation*, volume 535 of *Studies in Computational Intelligence*, pages 61–83. Springer, 2014.

[4] T. Baarslag, M. J. C. Hendrikx, K. V. Hindriks, and C. M. Jonker. Learning about the opponent in automated bilateral negotiation: a comprehensive survey of opponent modeling techniques. *Autonomous Agents and Multi-Agent Systems*, 30(5):849–898, 2016.

[5] V. P. Crawford and J. Sobel. Strategic information transmission. *Econometrica*, 50(6):1431–1451, 1982.

[6] E. Kamenica and M. Gentzkow. Bayesian persuasion. *American Economic Review*, 101(6):2590–2615, 2011.

[7] R. B. Myerson and M. A. Satterthwaite. Efficient mechanisms for bilateral trading. *Journal of Economic Theory*, 29(2):265–281, 1983.

[8] J. Ochs and A. E. Roth. An experimental study of sequential bargaining. *American Economic Review*, 79(3):355–384, 1989.

[9] A. Rubinstein. Perfect equilibrium in a bargaining model. *Econometrica*, 50(1):97–109, 1982.

[10] E. Shapira, O. Madmon, I. Reinman, S. J. Amouyal, R. Reichart, and M. Tennenholtz. GLEE: A unified framework and benchmark for language-based economic environments. *arXiv:2410.05254*, 2024. https://arxiv.org/abs/2410.05254

[11] R. Agarwal, M. Schwarzer, P. S. Castro, A. C. Courville, and M. G. Bellemare. Deep reinforcement learning at the edge of the statistical precipice. *Advances in Neural Information Processing Systems*, 34:29304–29320, 2021. https://proceedings.neurips.cc/paper/2021/hash/f514cec81cb148559cf475e7426eed5e-Abstract.html

A Supplementary behavioral evidence

Table A1: Buyer-template history and high-quality rate on the next purchase (source cohort: 6,185 buyer replays). Rates are conditioned on purchases and template history; cohort size is not the event count in each row. The all-low three-times estimate is below every configured immediate break-even threshold $1/m$, motivating a veto heuristic rather than proving dominance.

Prior record of template	hit rate	Prior record	hit rate
no prior	70.4%	all-low $\times 1$	33.8%
all-high $\times 1$	81.3%	all-low $\times 2$	19.3%
all-high $\times 2$	84.8%	all-low $\times 3$	5.3%
all-high $\times 3$	91.2%	mixed	63.7%

Table A2: Development-fleet volume, 12–30 August 2026, recovered from local turn logs. These are fleet totals, not the games of the single ranked slot.

Family	unique games	Fleet total	
Bargaining	89,641	logged turns	3,114,356
Negotiation	76,490	unique games	264,432
Persuasion	98,301	reconstructable runs	49

Table A3: Exploratory break-even diagnostics from 7,104 known-horizon and 31,208 open-horizon bargaining games. A fitted zero crossing is conditional on the observed rating process; it is not generally a field median. A separate 26,773-decision replay cohort had 98.79% control fidelity and exposed a gate as low as 0.118 in the known-horizon player-1 regime. It is distinct from the exact-replay cohorts reported in the main text. Differences are computed before rounding.

Horizon	Seat	Mean banked	Fitted zero crossing	Difference
open	p_1	0.530	0.476	$+0.054$
open	p_2	0.501	0.463	$+0.038$
known 12	p_1	0.435	0.816	-0.381
known 12	p_2	0.687	0.449	$+0.239$

Table A4: Selected policy changes and their diagnostic evidence. Evidence types and cohorts differ: live comparisons, local replay and transcript reclassification do not establish a common replication standard or game-theoretic dominance. A standardized effect is an exploratory effect-to-noise ratio, not a confirmatory significance level.

Change	Evidence
Complete-info ultimatum	Larger demand in a one-round known-surplus seat; standardized effect $+3.42$, $n = 214$. Acceptance remains an opponent decision.
Rung-aware pricing	Two live comparisons with standardized effects $+3.9$ and $+4.1$.
Free-clock accept floor	Mean gain $+0.0458$ of pot over 108 refusals in a 1,285-game replay, standardized effect 7.7, with none worse. No discounting does not eliminate termination or rejection risk.
Open-horizon cap response	Replay identified 11 of 11 examined zero-payoff games affected by the termination-limit failure.
Negative-signal parser	270 purchases reclassified; 258 (95.6%) low quality, versus a 27.5% overall purchase-conditioned low-quality rate.

Table A5: Intervention register, from the change register kept during the competition and the project notes. It is a historical record, not a validated ledger: evidence types, units and cohorts differ, no entry was pre-registered, and the actions an intervention changed were counted only where replay is listed. Standardized effects are exploratory effect-to-noise ratios.

Intervention	Evidence (unit, cohort)	Recorded result	Status
Bargaining			
Round-cap accept collapse	Logs, then replay of a run-37 arm	Caught 11 of 11 games that would have hit the zero-payoff cap; the cap was later observed at round 96, not 99	shipped
Final-proposer seat price	Full record, 1,415 games	91.9% signed; seat worth 0.507 of pot, not 0.552	shipped
Accept slack 0.97 to 0.99	Simulator, 14,000 games	Inert within the simulator spread	shipped
Free-clock accept floor	Replay, runs 41–42, 1,285 games	Mean $+0.0458$ of pot over 108 refusals, none worse	shipped
Settle-early bundle	Unbundled replay	Whole gain came from the accept floor; other knobs added nothing	dropped
Bounded-only accept bar	Live arm, then control	$+2.20$ and $+1.61$ became -0.015 and $+0.026$ of pot	reverted
Lower opening demand	9,370 games, within-cell	Pays only where a refused round is costly	unchanged
Free-delay holds and cap	Simulator; run 37	Gate opened twice in five hours	unmeasured
Counter to sweepers	Run 39, 14 minutes	Drained before a measurement	unmeasured
Discounted hold cap	389 historical refusals	Refusing cost 0.0464 of pot	built, off
Negotiation			
Rung-aware pricing	Two live comparisons	Standardized effects $+3.9$, $+4.1$	shipped
Complete-info ultimatum	Live arm, 214 games	Standardized effect $+3.42$	shipped
Stonewall holds to endgame	Run 37, paired within game, 23 games	12 better, 11 equal, 0 worse	shipped
Untradable-seat walk-away	One slot's logs, 26–30 Aug	Walked all 1,380 seats with a decision turn	shipped
Late concession, complete info	Replay	Changes 25.5% of bounded complete-info decisions	armed
Hide rung from last word	Run 37	Close rate halved to 4.2%; premise was a survivorship artifact	off
Midpoint rail	Construction	Cannot bind given the ladder	removed
Stonewall respects ultimatum	Construction	Unreachable condition	dead code
Known-zone floor	Replay, 13,056 decisions	Changed 0 decisions	inert
Rung shade 0.30–0.50	Fleet-wide, no control	Ratings fell 100–385 points	reverted
Persuasion			
Negative-signal parser	Re-scoring, 123,700 buyer rounds	270 purchases reclassified, 258 low quality	shipped
Hard-regime rationing	Replay, 13,479 seller turns	Changed 0 decisions	inert
Low-quality push to 100%	29 agent-runs, centred within run	Slope -0.048 (s.e. 0.162)	not shipped
Lie budget 0.90 and 1.00	Live arms	The lever moved less than the control spread	null
"Buyers punish lies"	Observational	Explained by purchase selection; buyer responses not separately identified	retracted
Other tuning arms (live arms against simultaneous controls)			
Patience floor	Live arm	Clamped downstream to a 2-point move	null
Accept bar, two arms	Live arms	Within the control spread	null
Stall clock	Live arm	Within the control spread; counts not recorded	null
Anchor grace	Live arm	Within the control spread; counts not recorded	null
Bargaining messages off	Live arm	Within the control spread; counts not recorded	null
Last-word demand	Live probe	Acceptance flat: 94.8% at 0.97 against 90.2% at 0.90	null

Table A6: Selected complete-information seller cells from a 10,740-game negotiation analysis. The displayed seller cohorts contain 957 games; the 469-game horizon-10 cohort is partitioned below. The association between late settlement and low payoff does not identify the causal effect of waiting.

Cell / outcome	n	Payoff measure	rating / game
$\mathrm{max_rounds} = 1$	488	asks 0.991 of surplus; 99.2% sign	$+6.10$
$\mathrm{max_rounds} = 10$	469	partitioned below	-4.63
closed by round 8	66	banked / buyer value: 0.2365	$+5.9$
agreed in round 10	403	banked / buyer value: 0.0433	-6.4

B Additional methods and limits

Exploratory break-even fits. Regressing rating delta on banked payoff within a cell supplied diagnostic zero crossings (Table A3). Slopes were positive in 58 of 64 cells, with median $R^2 = 0.60$ and split-half correlation $r = 0.741$. These describe regularity in the logged sample. A root depends on rating history, opponent adjustments and the reference pool; it is not generally a recovered field median.

Aggregate inference for declined products. Let p be the prior probability of high quality in a persuasion cell, b the purchased fraction, and h the observed high-quality rate among purchases. The population identity $p = bh + (1 - b)q_{\mathrm{declined}}$ motivates

$$\widehat{q}_{\mathrm{declined}} = \frac{p - \widehat{b}\,\widehat{h}}{1 - \widehat{b}}, \qquad \widehat{b} < 1.$$

This uses the prior in expectation, not the realized total number of high-quality units in a finite sample. It estimates an aggregate rate and cannot recover the quality of a particular declined product. Near-complete purchasing makes the estimate unstable; sampling variation can place it outside $[0, 1]$. At $b = 0$ it supplies no information beyond the prior. It should not be used as evidence about a single hidden round or as a precision estimate without uncertainty.

Simulation and memory. A local bargaining simulator fitted to 15,788 games and eight adversarial archetypes reproduced per-cell median settlement rounds but overestimated payoff by approximately 0.03 of the pot. It was used for candidate screening rather than as evidence of a live treatment effect. Opponent profiles showed split-half correlation $r = 0.786$, but only 19 opponent-by-cell profiles had at least 25 purchases in each half. Identity was withheld in 49.9% of games, and no quality was revealed in 42,616 unbought rounds. Purchase selection and sparse profiles limit the conclusions for cross-game opponent modeling [4].

Operational versioning. Experimental flags written into a working tree, entry points outside a copied package and abrupt mid-game termination can make intended and deployed behavior differ. A reference build passing tests does not establish that every concurrent slot ran that build. In the identical-policy experiment of Section 3, all five slots were deployed from one reference build by a script that sets every flag explicitly and checks it from inside each slot's own directory before launch. Per-slot source, configuration and entry-point hashes were not recorded, so slot identity rests on that procedure rather than on per-slot verification. Exact replay is tied to the logged cohort and policy version.

C Aggregation and outcome-dependent selection

A candidate rule used whether a seller had previously declined an opportunity to sell. Pooled across fifteen (m, p) cells, later low-quality rates were 0.285 / 0.287 / 0.286 / 0.315 for sellers declining 0 / 1 / 2 / 3–4 of their first four opportunities. Within $(m = 1.20, p = 0.80)$, the corresponding values were 0.160 / 0.102 / 0.081 / 0.069 (cell sample $n = 9,932$), against an immediate break-even low-quality rate $1 - 1/m = 0.167$. This illustrates how configuration mixture can mask a within-cell pattern; it is not evidence of a uniform reversal in every cell. The rule was not deployed.

Separate analyses were affected by outcome-dependent inclusion: deal rates computed only among closed deals, quality estimates conditional on purchases, and abandonment estimates from logs missing a completed final turn. These are selection problems, distinct from aggregation over cells.

One persuasion arm also changed its standardized effect from -3.11 to -0.82 to $+3.02$ when the minimum-count filter admitted 3, 33 or 44 cells. This sensitivity motivates specifying inclusion rules before inspecting outcomes and treating retrospective screening statistics as exploratory.

D Rating-model and reproducibility boundaries

An idealized independent two-arm comparison with equal variance σ^2 and equal sample size n has standard error $\sigma\sqrt{2/n}$. Requiring an effect Δ to equal three standard errors gives $n = 18\sigma^2/\Delta^2$. Earlier screening applied this arithmetic to per-game rating deltas, whose observed standard deviations were 4.84 / 4.42 / 4.44 over 120,023 joined games. At $\Delta = 0.25$, it yields roughly 5,600–6,700 observations per arm using unrounded estimates. This is an IID sensitivity illustration, not a validated policy-strength detection bound: deltas depend on current ratings and telescope over a run, and the criterion does not specify statistical power. Fixed-reference game scores or properly reconstructed game targets, with time and configuration controls, would provide a better-defined comparison.

Similarly, a fitted zero rating-delta crossing is not generally the median of the current payoff distribution. Rating history, opponent strength and a changing reference pool enter that relationship. Comparisons with equilibrium predictions additionally require the correct initial seat and horizon; an infinite-horizon proposer share must not be treated as the equilibrium payoff for all seats and finite horizons.

For a constant-η EMA, perturbing one game target by d changes the rating by $\eta d(1-\eta)^k$ after k later games if those later targets remain fixed. The recorded sum of abandonment deltas therefore cannot be interpreted as a final-standing effect. A persistent stream of abandonments and a finite cluster near stopping time are different estimands; no precise persistent-cost estimate is claimed here.

The public release includes policy code, tests and analysis scripts. Its scripts require joined game records and human-history exports that are not distributed alongside the code because they include opponents' transcripts. Organizers have indicated plans to release game records, but local deployment flags, rating-poll snapshots, filter choices and joins are additional provenance requirements. The released method is inspectable; the empirical results are not yet a one-command reproduction. Tables retain their analysis-specific cohorts, and aggregate counts should not be substituted for the denominators of individual comparisons.

For the identical-policy experiment, a regenerable record needs, per snapshot and slot, a timestamp, the overall and three family ratings, cumulative games by family and role, and the deployed source, configuration and entry-point hashes. The camera-ready release adds the 45 plotted overall ratings and each slot's game counter and rating at the first and last snapshot (`paper/camera_ready/data`). The family-level snapshots behind the within-family spreads are not part of the release, and deployment hashes were not recorded. The server's exact update rule was not available to me; the API reports each game's rating change, from which a target g_t can be backed out only under an assumed η. Matching the five slots on configuration, role and opponent strength would need every slot's per-game records, and the turn logs I retained cover one slot, so the comparison in Section 3 stays descriptive.

Cohorts. Analyses draw on different records, and a count from one should not serve as a denominator for another: 264,432 unique games in the fleet's turn logs (Table A2); 120,023 games joined to their rating deltas (the delta calculations above); 109,881 rated games in the account's history feed, which also covers games without a completed final turn (abandonment); 6,185 buyer replays holding 123,700 rounds (parser re-scoring and Table A1); and the persuasion purchases behind Table 1.